Rachana Gurudu

480.694.5064 | rachana.gurudu@case.edu |
https://www.linkedin.com/in/rachana-gurudu

EDUCATION

Case Western Reserve University (CWRU) Cleveland, OH

M.D. Candidate *Expected 05/2031*

- Pre-Professional Scholar in Medicine (8-Year BA/MD)

M.S. in Biomedical and Health Informatics *Expected 05/2027*

- GPA: 4.0/4.0
- Relevant Coursework: AI in Medicine: Knowledge Representation and Deep Learning

B.A. in Chemical Biology and Healthcare Management *Expected 05/2027*

- *Minor:* Medical Anthropology
- *GPA:* 4.0/4.0
- Relevant Coursework: Organic Chemistry, Biochemistry, Analytical Chemistry, Health Economics, Biology, Physics

RESEARCH

Cleveland Clinic *Cleveland, OH*

Research Intern, Mentors: Dr. Animesh Tandon, Dr. Po-Hao Chen *01/2024 - Present*

- Studying view classification methods of cardiac MRI and assessing multimodal AI technologies for the monitoring and treatment of pediatric patients with congenital heart disease
- Drafting review paper analyzing generative AI in radiology education
- Previously investigated epigenetic mechanisms of neuroendocrine bladder cancer in the Mian Lab

CWRU Department of Chemistry *Cleveland, OH*

Student Researcher, Shane Parker Lab *08/2025 - Present*

- Simulating stereoisomeric conformations of organic molecules using Conformer Ensemble Screening and Optimization (CENSO) in a high performance computing environment

CWRU Clinical and Translational Science Collaborative *Cleveland, OH*

Student Researcher, Mentor: Dr. Clara Pelfrey, PhD *01/2025 - Present*

- Create low-cost AI-based evaluation systems for future leadership development programs
- Evaluate leadership outcomes of women faculty participating in the FLEX Leadership Development Program at CWRU School of Medicine

University Hospitals Cleveland Medical Center Cleveland, OH

Research Assistant, Mentor: Dr. Sree Harsha Tirumani, MD *04/2024 - Present*

- Analyze clinical incidence and interventions for MRI-related claustrophobia
- Evaluate the role of imaging core laboratories in academic cancer centers
- Assess stereotactic body radiation therapy for renal cell carcinoma

Mayo Clinic *Rochester, MN & Phoenix, AZ*

Clinical Research Intern, Mentor: Dr. Jeff Meeusen, PhD *05/2025 - 08/2025*

- Developed ML models to estimate low-density lipoprotein levels and glomerular filtration rates
- Surveyed patients on efficacy of at-home capillary blood collection devices
- Participate in clinical shadowing and seminars through the Clinical Research Internship Study Program (CRISP)

Research Intern - Department of Radiology, Mentor: Dr. Bhavika Patel, MD *12/2021 - 05/2023*

- Studied patient testing burden and testing morbidities index of Contrast Enhanced Mammography

Research Intern - Department of Gastroenterology, Mentor: Dr. Francisco Ramirez, MD *12/2021 - 01/2023*

- Studied prevalence of colorectal neoplasia in patients under 50 through retrospective chart review
- Presented findings at ACG 2022 and 2023, and received Outstanding Poster Award

Rachana Gurudu

480.694.5064 | rachana.gurudu@case.edu |
https://www.linkedin.com/in/rachana-gurudu

PROFESSIONAL EXPERIENCE

Cleveland Clinic Cleveland, OH

Patient Companion *04/2026 to 08/2026*

- Provide direct patient care to high-risk patients, monitoring clinical status and supporting nursing staff within an internal medicine and nutritional insufficiency unit

Case Western Reserve University Cleveland, OH

Peer Tutor and Undergraduate TA *01/2025 to Present*

- Conduct targeted review sessions, grade assessments, and provide individualized test-taking strategies for students across introductory biology, physiology, and medical anthropology courses

Orientation Leader *08/2024*

- Mentored a cohort of 20 incoming students, facilitating university programming, title IX and sexual violence training, and campus integration during Discover Days

Scottsdale Education Center Scottsdale, AZ

SAT/ACT Tutor 05/2023 to 08/2024

- Mentored students in preparation for SAT/ACT exams; Taught key concepts and test-taking strategies

COMMUNITY SERVICE & EXTRACURRICULARS

Case Western Reserve University - Disability Resources Cleveland, OH

Founder of Student Advisory Board, Peer Notetaker 01/2025 to Present

- Establish and lead a multidisciplinary student board to advise university administration on campus-wide advocacy, accessibility programming, and inclusive institutional policies.
- Remediate coursework to improve digital accessibility under University Technology

Universal Health Aid - Cleveland Cleveland, OH

President 08/2023 to 05/2026

- Direct an executive board to lead public health advocacy and community education initiatives, including bi-semester roundtables, campus events, and a community mental health database

Indian Classical Dance Cleveland, OH & Phoenix, AZ

President, CWRU Nritya 06/2011 to Present

- Direct weekly practices, design choreography, and lead a 8-person executive board and 40-member team to coordinate campus performances and regional competitions
- Arangetram Graduate: Completed a two-hour solo debut performance following 10 years of intensive training at the Arathi School of Dance

Phoenix Children's Hospital Phoenix, AZ

Float Volunteer 05/2024 to 08/2024

- Assist nurses, child life specialists, and physicians to enrich patient hospital experience through direct patient interaction with children in pediatric and cardiovascular ICU for 100 hours

Kappa Alpha Theta - Eta Pi Cleveland, OH

Vice President, Administration 01/2024 to 05/2026

- Manage chapter membership with regards to academic and service accountability, maintain and edit national organization bylaws, supervise chapter roster and involvement, and elect new leadership

PUBLICATIONS & PRESENTATIONS

1. Park et al (2026). Development and Validation of a Simplified Martin-Hopkins LDL-C Equation Using Machine Learning. *JAMA Cardiology.* https://doi.org/10.1001/jamacardio.2026.2314

Rachana Gurudu

480.694.5064 | rachana.gurudu@case.edu |
https://www.linkedin.com/in/rachana-gurudu

2. Swartchick CB, **Gurudu R**, Lexvold N, et al (2026). Feasibility of Capillary Self-Collected Blood Specimens for Routine Outpatient Laboratory Testing. *The Journal of Applied Laboratory Medicine*. https://doi.org/10.1093/jalm/jfag075
3. Pelfrey C**, Gurudu R,** Kusina J, et al (2026). Do Leadership Programs create future leaders? Using AI to Track Leadership Outcomes. OPEG Spring Exchange Conference, Columbus, Ohio.
4. **Gurudu, R**, Bansal, D, Chauhan, A, et al (2026). Tumor metrics imaging core labs: primer for radiologists. *Abdominal Radiology*. https://doi.org/10.1007/s00261-026-05439-8
5. **Gurudu R**, Lexvold N, Price L, et al (2025). Evaluation of capillary blood self-collection devices for EDTA whole blood: sample adequacy and user experience. Poster presented at the 2025 ADLM Preanalytical Phase Conference, Providence, Rhode Island. *Journal of Applied Laboratory Medicin*e. https://doi.org/10.1093/jalm/jfaf209
6. Naganatanahalli M, **Gurudu R**, Bhargava M, et al (2025). Revisiting MRI Claustrophobia: Incidence, Factors, Interventions. *Journal of Computer Assisted Tomography.* doi:10.1097/RCT.0000000000001806
7. **Gurudu R,** Woldea H, Lantz E, et al (2025). xGFR: A Machine Learning-Based Prediction Model to Improve Estimation of Glomerular Filtration Rate. *Mayo Clinic Clinical Research Internship Study Program Symposium*.
8. **Gurudu R**, Raju A, Ball S, Veizi E (2025). Patient Selection Criteria and Outcomes of Spinal Cord Stimulation for Treatment of Chronic Intractable Pain in Veteran Populations. *Intersections - Case Western Undergraduate Research Symposium*.
9. Zhao J, Mirak SA, **Gurudu R**, et al. (2024). Role of Radiologist in Gastroenteropancreatic Neuroendocrine Neoplasms Multi-Disciplinary Team. *Radiological Society of North America Annual Meeting*.
10. Patel, BK, **Gurudu, R**, Mazza, G, et al. (2023). Patient-Reported Testing Burden of Screening Contrast-Enhanced Mammography (CEM). *Journal of the American College of Radiology.* doi:10.1016/j.jacr.2023.11.009
11. Pisipati, S, **Gurudu, R**, Arguinchona, JH, et al. (2023). Estimating the Prevalence and Characteristics of Colorectal Neoplastic Lesions in Young Patients Undergoing Colonoscopy. *American College of Gastroenterology Annual Meeting*.
12. Pisipati, S, **Gurudu, R**, Arguinchona, JH, et al. (2022). Evaluating the Burden and Predictors of Colorectal Neoplastic Lesions in Young Patients Undergoing Diagnostic Colonoscopy. *American College of Gastroenterology Annual Meeting*.
13. **Gurudu, R** (2022). Comprehensive Bioinformatics Meta-Analysis of Coronary Artery Disease and Myocardial Infarction. *Research Archive of Rising Scholars*.

AWARDS & RECOGNITION

Dean's High Honors *Fall '23 - Spring '26*
U.S. Presidential Scholar *05/2023*
National Merit Scholar *05/2023*

SKILLS

- R and RStudio, Python, Machine Learning (TensorFlow and XGBoost), RedCap, Beginner Java and HTML, MS Excel and Office, Graphic Design (Canva), Intermediate Proficiency in Spanish